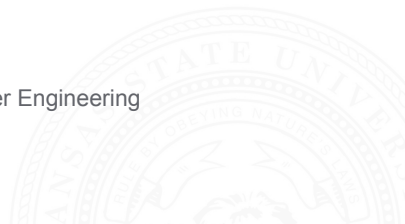


Transformer Modeling III

Determining model parameters from transformer tests

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Lecture outline

- 1 Part 1: Use the open-circuit (OC) test to determine excitation parameters
- 2 Part 2: Use the short-circuit (SC) test to determine series parameters
- 3 Part 3: Analyze voltage regulation and efficiency

L11 established the equivalent circuit. Today the OC and SC tests determine its unknown parameters, which then predict loaded operation.

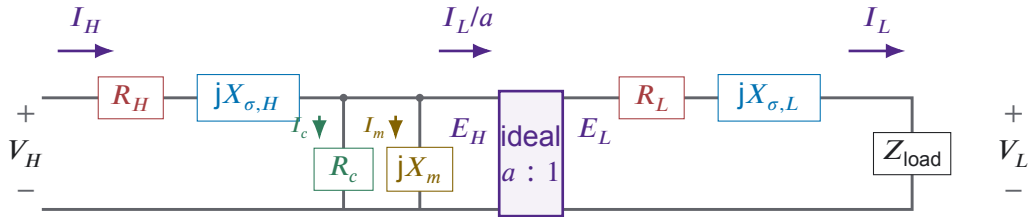
Determining model parameters

STARTING PROBLEM

We know the equivalent circuit of a practical transformer, but its parameter values are unknown. How can we determine these parameters experimentally?



The model established in L11



Known from L11

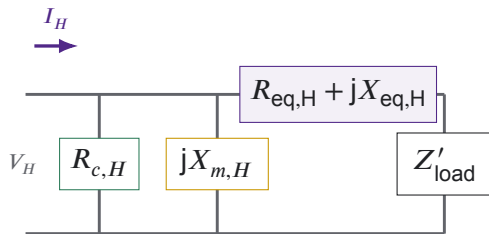
Circuit structure and the physical meaning of each element.

Still needed for a real transformer

Numerical winding and excitation parameters from measurements.

The complete model explains the test response. The experiments will estimate the parameters of its useful approximation.

Parameters to determine



Unknown parameter values

$$R_{c,H} = ?, \quad X_{m,H} = ?$$

$$R_{eq,H} = ?, \quad X_{eq,H} = ?$$

Approximate model from L11,
expressed on H.

$$R_{eq,H} = R_H + a^2 R_L, \quad X_{eq,H} = X_{\sigma,H} + a^2 X_{\sigma,L}.$$

The tests determine four parameters of this model. They do not determine each winding impedance separately.

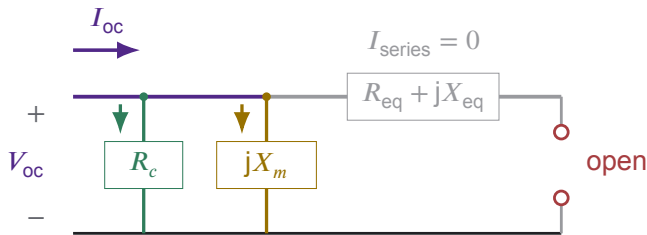
PART 1

Open-circuit test

Test condition, choice of winding, measurements, and excitation parameters

Open-circuit condition

Begin with the approximate circuit on page 5. Leave the opposite winding open.



Model viewed from the energized winding.

Follow the current

$$I_{load} = 0$$

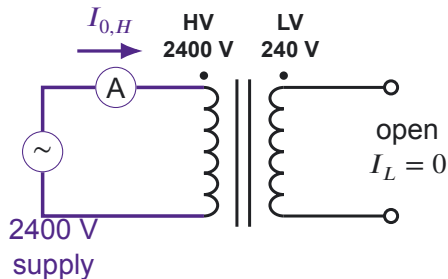
The open terminals stop the load current. The two excitation paths remain connected.

Apply rated voltage at rated frequency to observe core excitation.

The open-circuit test reveals R_c and X_m . Next choose which winding to energize and where to connect the meters.

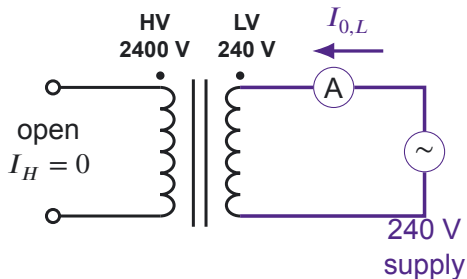
Why energize LV for the OC test?

Supply HV



$V_{\text{supply}} = 2400 \text{ V}$, measured current $I_{0,H}$

Supply LV: chosen here

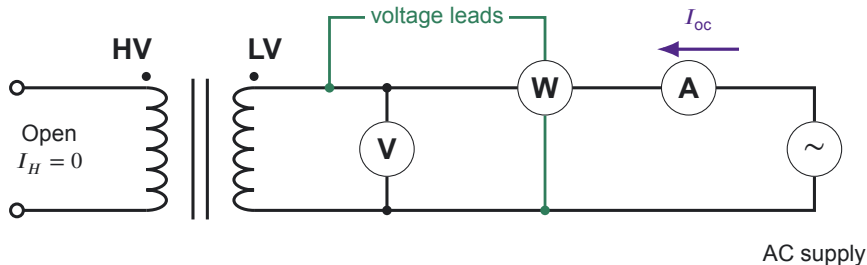


$V_{\text{supply}} = 240 \text{ V}$, $I_{0,L} \approx 10I_{0,H}$

Supplying LV needs one tenth the voltage and gives about ten times the no-load current to measure. Keep HV open.

Open-circuit test

Nameplate: 10 kVA, 2400/240 V (HV/LV).



Ammeter (A)

I_{oc} : RMS magnitude [A].

Voltmeter (V)

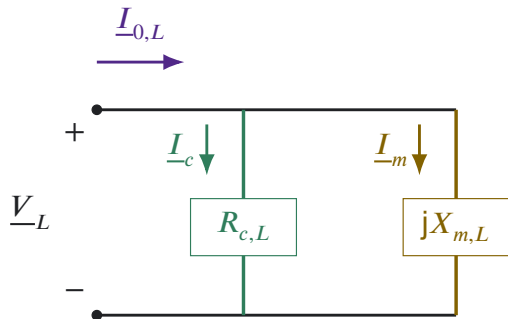
V_{oc} : RMS magnitude [V].

Wattmeter (W)

P_{oc} : active (real) power [W].

Keep HV open; apply 240 V RMS to LV. The three readings are real numbers; no meter here reports an angle.

Meter readings and RMS phasors



What the instruments report

$$V_{oc} = |\underline{V}_L| \quad [\text{V}]$$

$$I_{oc} = |\underline{I}_{0,L}| \quad [\text{A}]$$

$$P_{oc} = \text{Re}\{S_{oc}\} \quad [\text{W}]$$

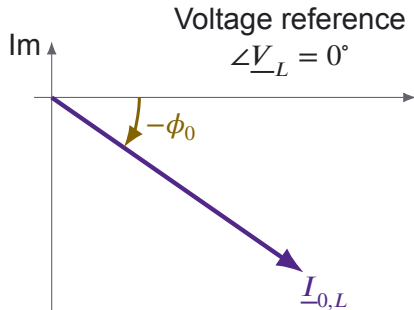
$$S_{oc} = \underline{V}_L \underline{I}_{0,L}^* = P_{oc} + jQ_{oc}$$

Sinusoidal RMS phasors are explicitly underlined.

L05–L06 convention: phasors are complex and use RMS magnitudes. The meters record magnitudes and real power.

Inferring the current phase

Use the sinusoidal approximation, as in L05–L06; neglect exciting-current harmonics.



$$\text{pf}_0 = \frac{P_{\text{oc}}}{V_{\text{oc}} I_{\text{oc}}} = \cos \phi_0$$

$$\underline{V}_L = V_{\text{oc}} \angle 0^\circ$$

$$\underline{I}_{0,L} = I_{\text{oc}} \angle (-\phi_0)$$

$$= I_{\text{oc}} (\cos \phi_0 - j \sin \phi_0)$$

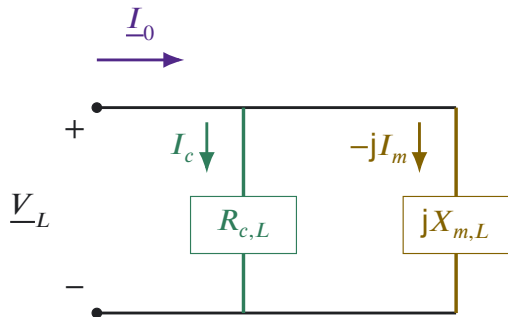
Polar and rectangular forms describe the same complex phasor.

Zero degrees is a reference choice.

The readings determine $\cos \phi_0$; the inductive model gives the lagging sign. No absolute phase is measured.

OC formulas: resolve the measured current

Neglect winding drops, copper loss, and exciting-current harmonics.



Follow the current split

$$\underline{I}_0 = I_c - jI_m, \quad I_{oc} = |\underline{I}_0|.$$

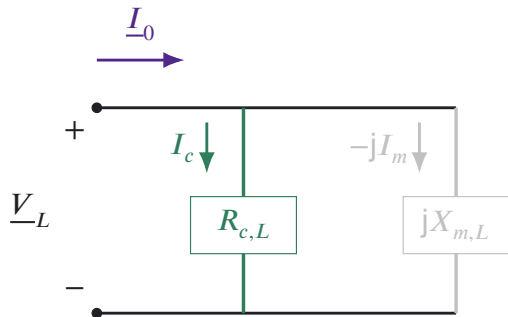
$$I_{oc}^2 = I_c^2 + I_m^2.$$

$I_c, I_m \geq 0$ are RMS component magnitudes. The branch phasors are I_c and $-jI_m$.

Voltage reference: $\underline{V}_L = V_{oc} \angle 0^\circ$.

The ammeter reads the magnitude of the phasor sum.

OC formulas: identify the core-loss branch



Wattmeter $\rightarrow$ core-loss current

$$P_{oc} \approx V_{oc} I_c$$

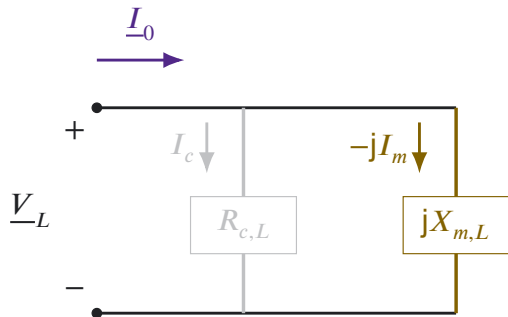
$$I_c = \frac{P_{oc}}{V_{oc}}$$

$$R_{c,L} = \frac{V_{oc}}{I_c} = \frac{V_{oc}^2}{P_{oc}}$$

Meter magnitude: $V_{oc} = |\underline{V}_L|$.

The inductive branch absorbs no average real power in this model; the wattmeter identifies $R_{c,L}$.

OC formulas: identify the magnetizing branch



Remaining current $\rightarrow$ reactance

$$I_m = \sqrt{I_{oc}^2 - I_c^2}$$

$$\underline{V}_L = (jX_{m,L})(-jI_m)$$

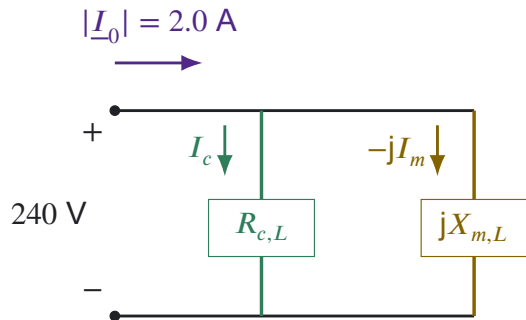
$$X_{m,L} = \frac{V_{oc}}{I_m}$$

Voltage reference: $\underline{V}_L = V_{oc} \angle 0^\circ$.

Both parameters are referred to the metered LV winding. Now apply these steps to the measured values.

Example: OC measurements

Same transformer: 10 kVA, 2400/240 V (HV/LV).



LV readings, HV open

$$V_{oc} = 240 \text{ V}$$

$$I_{oc} = 2.0 \text{ A}$$

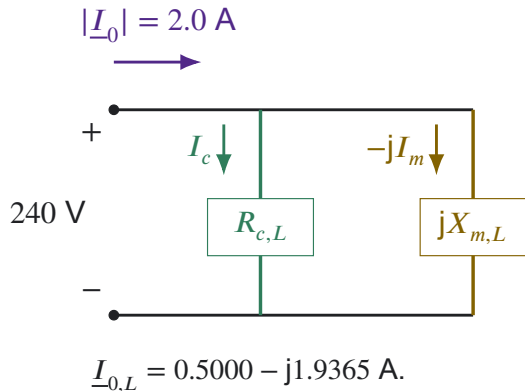
$$P_{oc} = 120 \text{ W}$$

Find the two shunt parameters.

Find $R_{c,L}$ and $X_{m,L}$ on LV.

Use measured power to find I_c , then measured current to find I_m .

Example: excitation-current components



$$I_c = \frac{120}{240} = \boxed{0.5000 \text{ A}}$$

$$I_m = \sqrt{2.0^2 - 0.5^2}$$

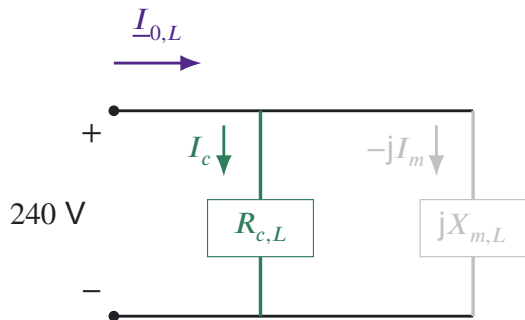
$$\boxed{I_m = 1.9365 \text{ A}}$$

$$\text{pf}_0 = 120/(240 \times 2) = 0.2500$$

$$\phi_0 = 75.522^\circ \text{ lagging.}$$

With $\underline{V}_L = 240\angle 0^\circ \text{ V}$, $\underline{I}_{0,L} = 2.0\angle(-75.522^\circ) \text{ A.}$

Example: core-loss resistance



Green branch: $I_c = 0.5000$ A.

$$R_{c,L} = \frac{V_{oc}}{I_c} = \frac{240}{0.5000}$$

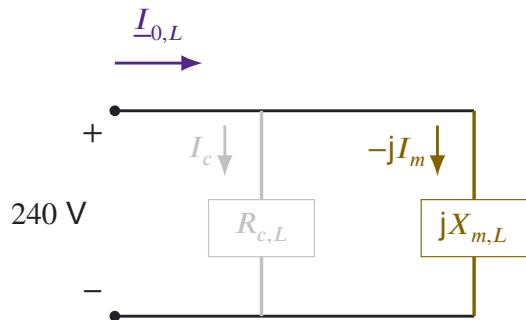
$$R_{c,L} = 480.0 \, \Omega$$

Check against the wattmeter:

$$\frac{240^2}{480.0} = 120 \text{ W.}$$

The OC wattmeter estimates core loss, represented by the green resistance.

Example: magnetizing reactance



$$X_{m,L} = \frac{V_{oc}}{I_m} = \frac{240}{1.9365}$$

$$X_{m,L} = 123.935 \, \Omega$$

Use the positive root for inductive reactance; the current lags voltage.

Gold branch: $I_m = 1.9365 \, \text{A}$.

The two shunt parameters are known. Next measure the remaining series winding parameters.

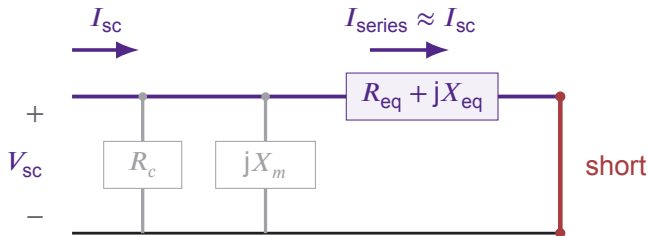
PART 2

Short-circuit test

Test condition, choice of winding, measurements, and series parameters

Short-circuit condition

The excitation parameters are known. Return to the remaining series elements on page 5.



Replace the load by a short connection.

Close the series path

The low series impedance allows a small applied voltage to drive rated winding current.

At this low voltage, excitation contributes little to the input current and power.

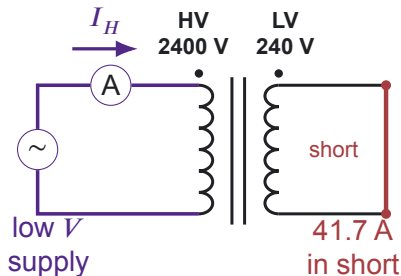
The short-circuit test reveals R_{eq} and X_{eq} . Choose the energized winding, then raise voltage only until rated current flows.

Why energize HV for the SC test?

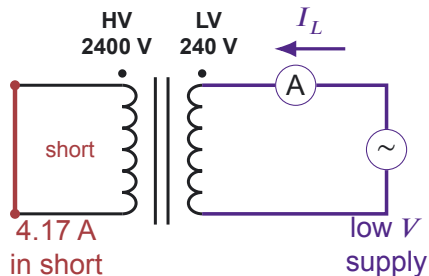
10 kVA transformer; winding voltages shown are nameplate ratings.

Supply HV: chosen here

Supply LV



$$\text{Source current: } I_H = \frac{10,000}{2400} \approx 4.17 \text{ A}$$

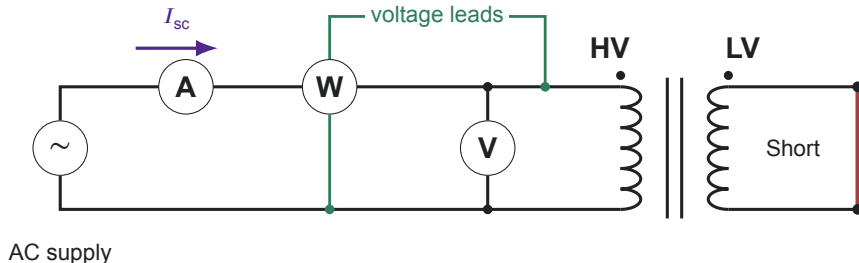


$$\text{Source current: } I_L = \frac{10,000}{240} \approx 41.7 \text{ A}$$

HV needs one tenth the source and instrument current. Raise the test voltage only until rated current flows.

Short-circuit test

Nameplate: 10 kVA, 2400/240 V (HV/LV).



Ammeter (A)

I_{sc} : RMS magnitude [A].

Voltmeter (V)

V_{sc} : RMS magnitude [V].

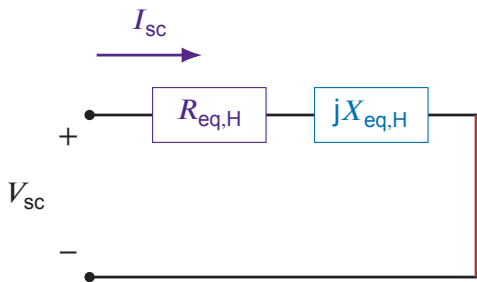
Wattmeter (W)

P_{sc} : active (real) power [W].

Short LV; raise the HV supply to rated RMS current. As in OC, record two RMS magnitudes and active power.

SC formulas: identify impedance and resistance

At reduced test voltage, excitation current and core loss are small.



$$|Z_{eq,H}| = \frac{V_{sc}}{I_{sc}}$$

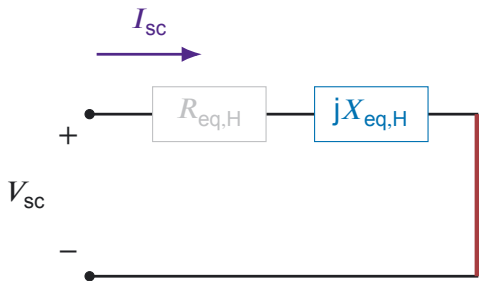
$$P_{sc} \approx I_{sc}^2 R_{eq,H}$$

$$R_{eq,H} = \frac{P_{sc}}{I_{sc}^2}$$

Low test voltage: neglect excitation.

The ratio of RMS magnitudes gives $|Z_{eq,H}|$. Real power identifies its resistive part.

SC formulas: identify leakage reactance



$$Z_{eq,H} = R_{eq,H} + jX_{eq,H}$$

$$|Z_{eq,H}|^2 = R_{eq,H}^2 + X_{eq,H}^2$$

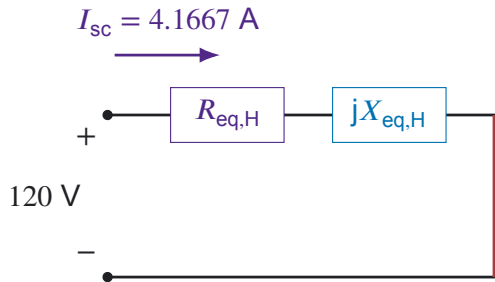
$$X_{eq,H} = \sqrt{|Z_{eq,H}|^2 - R_{eq,H}^2}$$

The blue element supplies the reactive voltage drop.

Take the positive root for leakage inductance. The test determines the combined series elements on HV.

Example: SC measurements

Same transformer: 10 kVA, 2400/240 V (HV/LV).



HV readings, LV shorted

$$V_{sc} = 120 \text{ V}$$

$$I_{sc} = 4.1667 \text{ A}$$

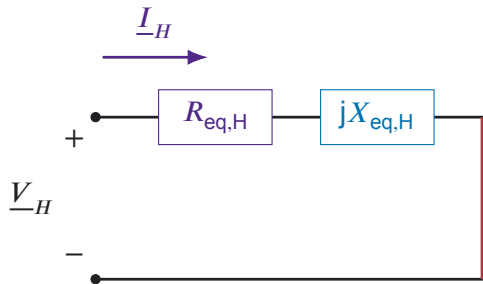
$$P_{sc} = 180 \text{ W}$$

Find $R_{eq,H}$ and $X_{eq,H}$.

HV-referred circuit with the load terminals shorted.

The series results already use HV. Keep OC results on LV until both parameter groups are available.

Example: SC current phasor



Rated RMS current: 4.1667 A.

$$\text{pf}_{sc} = \frac{180}{(120)(4.1667)} = 0.3600$$

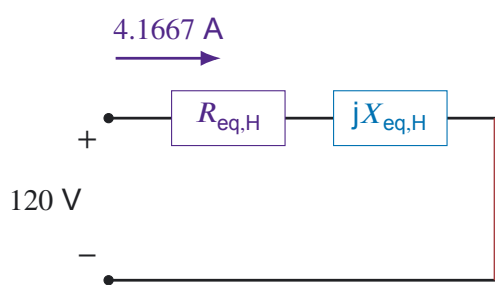
$$\phi_{sc} = \cos^{-1}(0.36) = 68.900^\circ$$

$$\underline{V}_H = 120 \angle 0^\circ \text{ V}$$

$$\underline{I}_H = 4.1667 \angle (-68.900^\circ) \text{ A}$$

The meters supply magnitudes and watts; the inductive series model supplies the lagging sign.

Example: series impedance magnitude

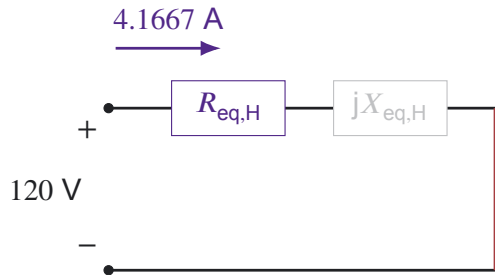


Voltage spans both series elements.

$$\begin{aligned} |Z_{eq,H}| &= \frac{V_{sc}}{I_{sc}} \\ &= \frac{120}{4.1667} = \boxed{28.800 \, \Omega} \\ \frac{V_{sc}}{V_{H,rated}} &= \frac{120}{2400} = 5.00\%. \end{aligned}$$

Rated current occurs at five percent voltage. Next separate resistance from reactance.

Example: series resistance



$$P_{sc} \approx I_{sc}^2 R_{eq,H}$$

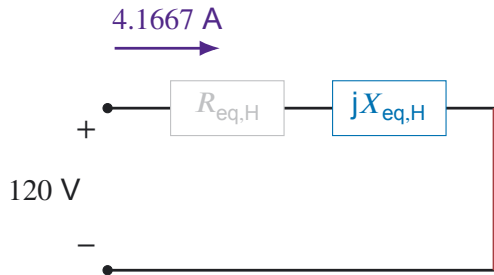
$$R_{eq,H} = \frac{180}{(4.1667)^2}$$

$$R_{eq,H} = 10.368 \, \Omega$$

The purple resistance accounts for P_{sc} .

At rated current, the SC wattmeter estimates full-load copper loss in both windings.

Example: leakage reactance



$$X_{eq,H} = \sqrt{|Z_{eq,H}|^2 - R_{eq,H}^2}$$

$$= \sqrt{28.800^2 - 10.368^2}$$

$$X_{eq,H} = 26.869 \, \Omega$$

The blue reactance represents combined leakage.

Both tests are complete. Refer their parameters to one side before assembling the model.

PART 3

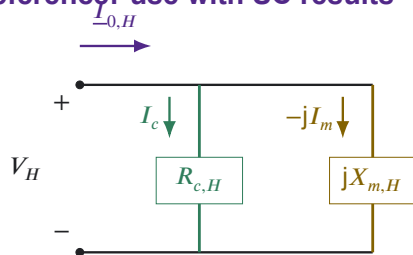
Voltage regulation and efficiency

Common reference side, load voltage, and real-power balance

Putting both test results on HV

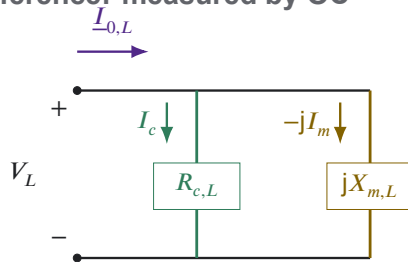
$a = 2400/240 = 10$: refer each LV shunt impedance to HV by multiplying by a^2 .

HV reference: use with SC results



$$R_{c,H} = 48.000 \text{ k}\Omega$$
$$X_{m,H} = 12.3935 \text{ k}\Omega$$

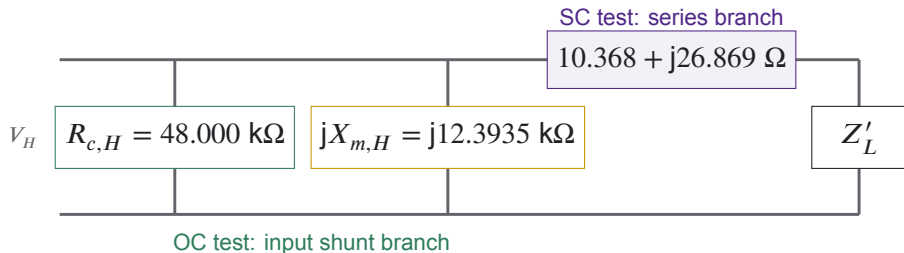
LV reference: measured by OC



$$R_{c,L} = 480.0 \text{ }\Omega$$
$$X_{m,L} = 123.935 \text{ }\Omega$$

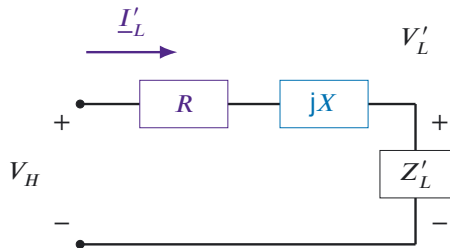
$R_{c,H} = a^2 R_{c,L}$ and $X_{m,H} = a^2 X_{m,L}$. The SC series values already use HV.

The L11 model with measured parameters



The original four unknowns now have measured estimates. Use this model to predict loaded operation.

Operation at rated load



$R = R_{eq,H}$, $X = X_{eq,H}$. All quantities are referred to HV.

Referred load at rated voltage

$$|S| = 10 \text{ kVA}, \quad \text{pf} = 0.80 \text{ lagging}$$

$$V'_L \approx 2400 \text{ V}$$

$$|I'_L| = \frac{10,000}{2400} = 4.1667 \text{ A}$$

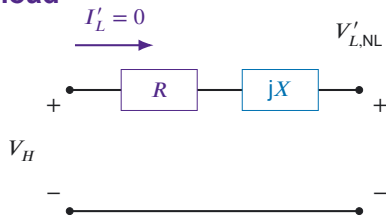
$$\underline{I}'_L = 4.1667 \angle -36.87^\circ \text{ A}$$

The load current through $R + jX$ produces the series voltage drop.

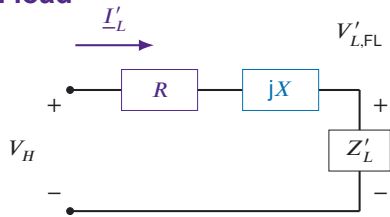
Voltage regulation: no-load and full-load

Keep the input voltage and frequency fixed; compare the load-terminal voltage.

No load



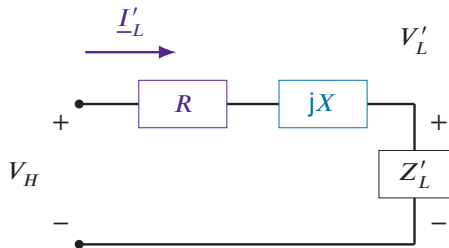
Full load



$$VR = \frac{|V'_{L,NL}| - |V'_{L,FL}|}{|V'_{L,FL}|} 100\%.$$

Neglect excitation for the voltage-drop estimate. With no load current, the series drop vanishes.

Voltage regulation: series voltage drop



$$\underline{V}_H \approx \underline{V}'_L + \underline{I}'_L(R + jX)$$

Component along the load voltage:

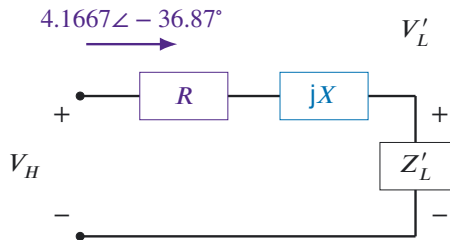
$$\Delta V \approx |I'_L|(R \cos \phi + X \sin \phi)$$

$$\text{VR} \approx \frac{\Delta V}{2400} 100\%$$

$R = R_{\text{eq,H}}$, $X = X_{\text{eq,H}}$. All quantities are referred to HV.

This is a first-order estimate at rated voltage. For leading current, the reactive contribution changes sign.

Example: voltage regulation



$R = 10.368 \, \Omega$, $X = 26.869 \, \Omega$ on HV.

Project the two series drops:

$$|I'_L| R \cos \phi = 34.560 \, \text{V}$$

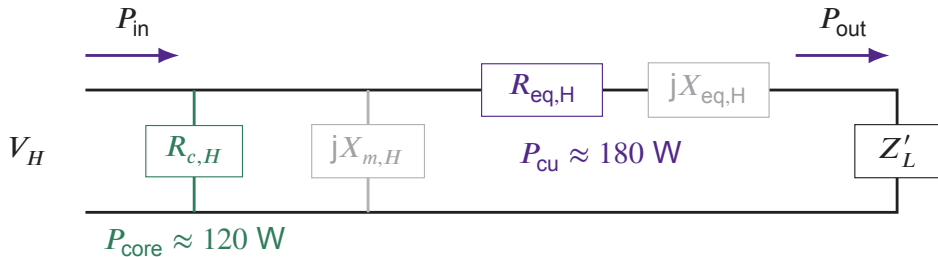
$$|I'_L| X \sin \phi = 67.173 \, \text{V}$$

$$\Delta V \approx 101.733 \, \text{V}$$

$$\text{VR} \approx 4.2389\%$$

The LV drop is about 10.173 V; dividing by 240 V gives the same regulation.

Losses in the operating circuit



Core loss: shunt resistance

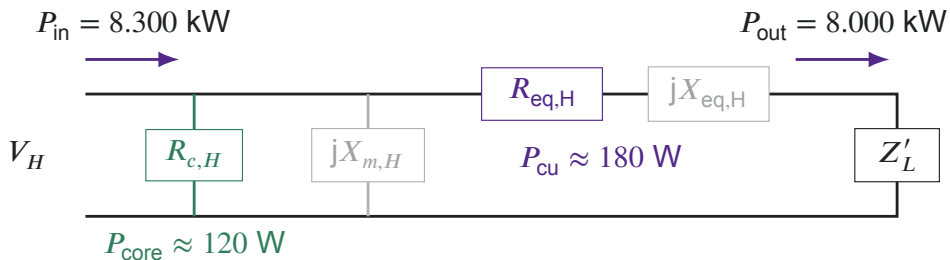
$$P_{\text{core}} \approx \frac{V_H^2}{R_{c,H}} = 120 \text{ W}$$

Copper loss: series resistance

$$P_{\text{cu}} \approx |I'_L|^2 R_{\text{eq},H} = 180 \text{ W}$$

At load fraction x , copper loss is about $180x^2 \text{ W}$; core loss is nearly constant at rated voltage.

Efficiency: real-power balance



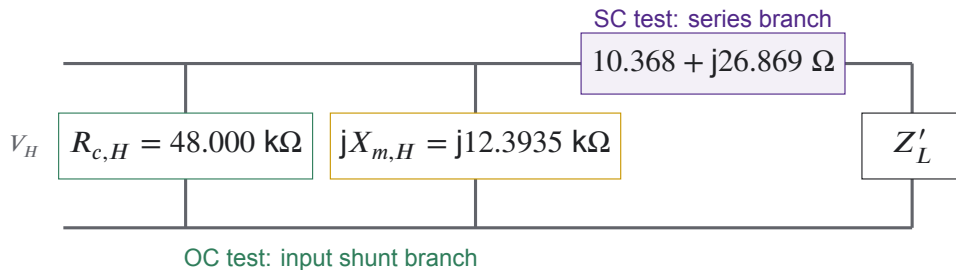
$$P_{out} = |S| \text{ pf} = 10(0.80) = 8.000 \text{ kW}$$

$$P_{in} \approx 8.000 + 0.120 + 0.180$$

$$\begin{aligned} \eta &= \frac{P_{out}}{P_{in}} 100\% \\ &= \frac{8.000}{8.300} (100\%) = \boxed{96.3855\%} \end{aligned}$$

Real-power input equals output plus core loss and copper loss. Use kW in the balance.

Summary: tests, regulation, and efficiency



Part 1: Open-circuit test

Rated voltage
 R_c and X_m

Part 2: Short-circuit test

Rated current
 R_{eq} and X_{eq}

Part 3: Regulation and efficiency

At rated load, 0.80 lagging pf:
 $VR \approx 4.24\%$, $\eta \approx 96.39\%$

OC and SC determine the circuit parameters; voltage drop and real-power balance predict performance.